Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 39)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following best describes a **representative** sample? Tick one box.

[1 mark]

A A sample taken from the most convenient location ☐

B A sub-sample that accurately reflects the complete data set ☐

C The largest sample that can be collected ☐

D A sample that supports the hypothesis ☐

0 1 . 2 Describe how random sampling sites would be selected in an irregularly shaped meadow.

[2 marks]

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0 1 . 3 Explain why deliberately choosing sampling sites introduces bias.

[2 marks]

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0 2 A pesticide was applied to a field of oilseed rape. Table 1 shows the concentration of the pesticide in organisms sampled from a pond next to the field.

Table 1

Organism	Trophic level	Pesticide concentration / mg kg^{-1}
Pond water	-	0.002
Phytoplankton	1	0.04
Water fleas (Daphnia)	2	0.5
Small fish	3	4.2
Heron	4	38.0

0 2 . 1 Calculate how many times more concentrated the pesticide is in the heron than in the pond water. Give your answer to two significant figures.

[2 marks]

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Answer [ ] times

0 2 . 2 Name the process shown by the data in Table 1.

[1 mark]

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0 2 . 3 Explain the **two** properties of the pesticide that cause the pattern shown in Table 1.

[4 marks]

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0 2 . 4 The pesticide was probably an organochlorine. Suggest why the use of organochlorines is now banned or heavily restricted in most countries, even though DDT is still used to control malaria mosquitoes in some houses.

[3 marks]

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0 3 Table 1 shows the pH of rainwater and the number of fish species in lakes in four areas of Scandinavia with different bedrock geology.

Table 1

Lake area	Bedrock	Mean rainfall pH	Mean lake pH	Number of fish species	Dissolved aluminium / mg dm^{-3}
A	Granite	4.3	4.6	0	0.42
B	Granite	4.8	5.1	3	0.20
C	Limestone	4.3	7.2	8	0.01
D	Limestone	5.5	7.6	9	0.01

0 3 . 1 State the pH below which rain is described as acid rain and explain why unpolluted rain is already slightly acidic.

[2 marks]

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0 3 . 2 Use Table 1 to explain why lakes in area C contain many more fish species than lakes in area A, even though the rainfall has the same pH.

[4 marks]

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0 3 . 3 Describe **two** effects of acid rain on non-living objects.

[2 marks]

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0 3 . 4 Explain how sulfur dioxide released from a power station is converted to sulfuric acid in the atmosphere.

[2 marks]

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0 4 . 1 Complete Table 1 to show the source and disposal method for each category of radioactive waste from nuclear electricity generation.

[6 marks]

Table 1

Waste category	Example / source	Storage or disposal method	
High-level waste			
Intermediate-level waste			
Low-level solid waste			

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0 4 . 2 Explain why high-level waste must be cooled and why its storage must be monitored for very long periods.

[2 marks]

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0 4 . 3 Describe how low-level liquid and gaseous wastes are treated before discharge.

[2 marks]

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0 5 . 1 Explain how integrated manufacture saves energy in the iron and steel industry.

[2 marks]

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0 5 . 2 Explain why recycling is not always energy efficient.

[2 marks]

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0 5 . 3 Explain how each of the following manages electricity infrastructure to conserve energy.

[4 marks]

High-voltage grid .....

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ICT management of supply and demand .....

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0 5 . 4 Give **two** locational factors affecting where new generating infrastructure is built.

[2 marks]

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0 6 The Aral Sea in central Asia was once the fourth largest lake in the world. From the 1960s water from the two rivers feeding it (the Amu Darya and Syr Darya) was diverted to irrigate cotton and rice. Table 1 shows changes in the lake.

Table 1

Year	**1960**	**1980**	**2000**	**2015**
Lake area / km^2	68 000	52 000	24 000	7 000
Salinity / g dm^{-3}	10	18	45	over 100
Commercial fish catch / thousand tonnes	40	10	0	0

0 6 . 1 Calculate the percentage reduction in the area of the Aral Sea between 1960 and 2015.

[2 marks]

Answer ☐ %

0 6 . 2 Explain why the salinity of the lake increased as the area fell.

[2 marks]

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0 6 . 3 Explain why the fish catch fell to zero, and describe **two** other environmental consequences of the shrinking of the lake.

[4 marks]

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0 6 . 4 Suggest why disagreements over the rivers continue between the countries of the region.

[2 marks]

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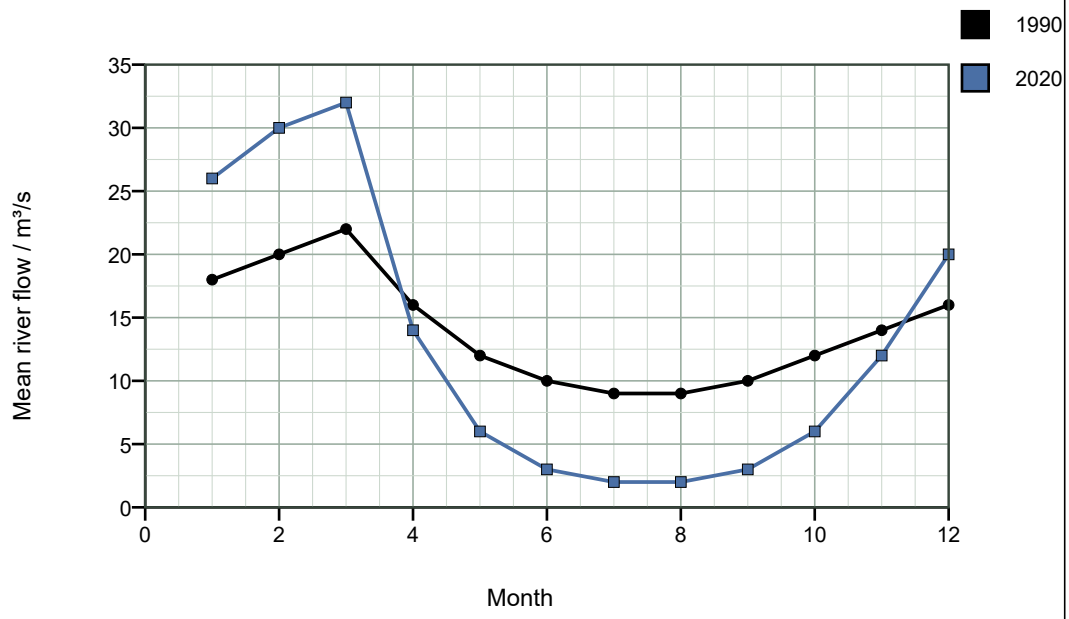
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0 7 Figure 1 shows the monthly flow of a river in South America that is fed partly by glacier meltwater, in 1990 and 2020. The glacier feeding the river shrank considerably between the two dates.

Figure 1



0 7 . 1 Compare the river flow in 1990 and 2020.

[3 marks]

0 7 . 2 Explain how the shrinking of the glacier explains the differences.

[3 marks]

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0 7 . 3 Suggest **two** problems the change in flow causes for people living downstream.

[2 marks]

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0 7 . 4 Explain why a glacier can be extending down its valley yet still be losing ice volume.

[2 marks]

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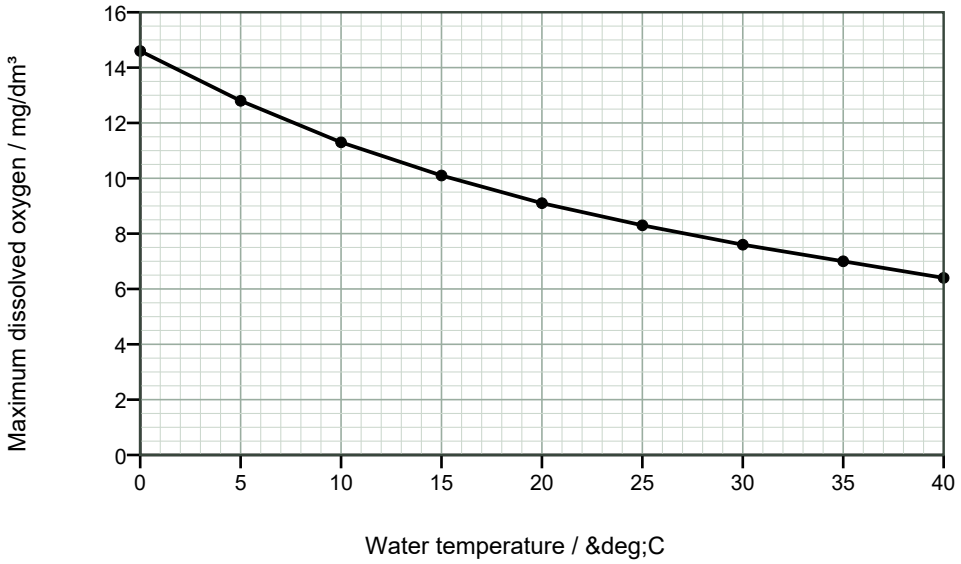
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0 8 Figure 1 shows the maximum concentration of dissolved oxygen in fresh water at different temperatures. A power station discharges warm cooling water into a river. The lowest dissolved oxygen concentration that trout can tolerate is 6.5 mg dm^{-3}.

Figure 1



0 8 . 1 Describe the relationship shown in Figure 1.

[1 mark]

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0 8 . 2 Use Figure 1 to determine the highest water temperature at which trout could survive if the water were fully saturated with oxygen.

[1 mark]

Answer °C

0 8 . 3 In summer the river upstream of the power station is at 20 °C and the effluent raises the temperature by 7 °C. Use Figure 1 to calculate the reduction in maximum dissolved oxygen concentration caused by the effluent.

[2 marks]

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Answer [ ] mg/dm³

0 8 . 4 Explain why the actual dissolved oxygen concentration downstream may be much lower than the values in Figure 1.

[2 marks]

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0 8 . 5 Describe **three** effects of warmer water on aquatic organisms other than reduced dissolved oxygen.

[3 marks]

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0 8 . 6 Describe how a cooling tower reduces the temperature of power station effluent water.

[1 mark]

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0 9 Figure 1 shows a simplified nitrogen cycle with some processes labelled 1 to 6.

Figure 1

Number	Process description
1	Nitrogen gas converted to ammonium ions by Rhizobium in root nodules
2	Ammonium ions oxidised to nitrite then nitrate by Nitrosomonas and Nitrobacter
3	Nitrates reduced to nitrogen gas by Pseudomonas in anaerobic soil
4	Amino groups in dead organic matter released as ammonium ions by decomposers
5	Nitrogen and oxygen react during lightning to form NOx
6	Nitrates washed out of soil into rivers

0 9 . 1 Name processes 1 to 6.

[3 marks]

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0 9 . 2 Explain why draining a waterlogged field increases the nitrate available to crops.

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0 9 . 3 Explain why nitrogen is described as a resource that is 'not very abundant' for living organisms even though the atmosphere is 78% nitrogen.

[2 marks]

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0 9 . 4 Explain why growing legumes such as clover in a crop rotation reduces the need for artificial fertiliser.

[2 marks]

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1 0 . 1 As high-grade mineral deposits become depleted, discuss the strategies that could be used to secure future supplies of metals.

[9 marks]

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1 0 . 2 State what is meant by the **stock** (resource base) of a mineral.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

OR

1 1 . 2 Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS